

Digital Egypt Pioneers Initiative

Skills Dynamix (R4)

[Fixora Website](#)

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It has been a valuable opportunity to gain practical experience by developing the **Fixora** project and applying the knowledge and skills we have acquired throughout our studies. This project has helped us better understand the processes of planning, analyzing, designing, and developing a real-world web application.

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ABSTRACT

Home maintenance services have become an essential part of daily life, as people frequently require professional assistance for repairs, installations, and maintenance tasks. However, finding reliable technicians and managing service requests through traditional methods can be time-consuming and inefficient.

This project presents **Fixora**, a web-based home maintenance service platform designed to connect customers with qualified technicians in a simple, secure, and efficient manner. The system allows users to register and create personal accounts, browse available maintenance services, search for skilled technicians, and book service appointments through an intuitive interface.

The platform includes several main pages such as **Home, Services, About Us, Contact Us, Booking, Technician Profile, User Dashboard, and Login/Sign Up**. Customers can submit and track service requests, while technicians can manage appointments, update service status, and communicate with customers. Administrators can manage users, technicians, services, and bookings to ensure the smooth operation of the platform.

Overall, **Fixora** aims to improve the home maintenance experience by providing a user-friendly, efficient, and centralized platform that simplifies the process of requesting, managing, and delivering maintenance services.

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CHAPTER 1

INTRODUCTION

Chapter 1: Introduction.

In today's fast-paced world, finding reliable home maintenance and repair services has become a common challenge for many people. Whether it is plumbing, electrical work, carpentry, painting, or appliance repair, customers often struggle to find skilled technicians quickly and at reasonable prices. Traditional methods of searching for service providers through personal recommendations or social media are often time-consuming and lack transparency.

This project aims to develop a web-based **Fixora System**, an online platform that connects customers with professional technicians in a simple, efficient, and secure way. The system allows users to create personal accounts, browse available services, book appointments, communicate with technicians, and manage their service requests through an organized interface.

The platform includes several main pages, namely **Home, Services, About Us, Contact Us, Service Details, Booking, Technician Profile, User Dashboard, and Login/Sign Up**.

Through this platform, customers can request maintenance services, schedule appointments, track the status of their requests, and provide feedback after service completion. Technicians can manage their profiles, receive booking requests, update job statuses, and communicate with customers. The system also provides secure user authentication, ensuring personalized access for both customers and technicians.

The primary goal of this project is to provide a user-friendly, efficient, and centralized solution that improves the process of requesting and managing home maintenance services while enhancing communication between customers and service providers.

1.2 Project Idea and Scope

The **Fixora** project aims to develop a web-based Home Maintenance Management System that connects customers with qualified technicians through an integrated online platform. The system provides various maintenance services, including plumbing, electrical work, carpentry, painting, air conditioning maintenance, and appliance repair.

The platform supports user registration and authentication for both customers and technicians. Customers can browse services, search for nearby technicians, book appointments, monitor service requests, and rate completed services. Technicians can manage their profiles, receive and respond to booking requests, update service progress, and communicate with customers.

The system includes major pages such as **Home, Services, About Us, Contact Us, Service Details, Booking, Technician Profile, User Dashboard, and Login/Sign Up**. It is designed to provide a secure, responsive, and user-friendly web application that simplifies the process of requesting home maintenance services while ensuring reliability, convenience, and improved customer satisfaction.

1.3 Problem Definition

We talked to customers to identify the common problems they face when looking for home maintenance and repair services. We found the following issues:

1. Customers find it difficult to locate reliable and qualified technicians for different maintenance services through a single trusted platform.
2. Traditional methods of searching for technicians, such as personal recommendations or social media, are time-consuming and often unreliable.
3. There is a lack of a centralized system that allows customers to book, track, and manage maintenance service requests efficiently.
4. Communication between customers and technicians is often unorganized, leading to delays, misunderstandings, and missed appointments.
5. Customers have limited information about technicians' experience, availability, ratings, and service history, making it difficult to choose the right service provider.
6. Technicians lack an efficient platform to manage booking requests, update job status, and communicate with customers.
7. The absence of an integrated online system reduces convenience and makes the overall maintenance process less efficient.

1.4 System Objectives

1. Provide a unified platform for home maintenance and repair services.
2. Allow customers and technicians to register, log in, and manage their personal accounts securely.
3. Enable customers to browse different maintenance services and find suitable technicians.
4. Allow users to book, schedule, and manage service appointments easily.
5. Enable technicians to receive, accept, and manage service requests efficiently.
6. Provide technician profiles that include experience, availability, and customer ratings.
7. Allow customers to track the status of their service requests from booking to completion.
8. Improve communication between customers and technicians through an organized platform.
9. Provide a secure, user-friendly, and responsive web application.
10. Enhance convenience, efficiency, and customer satisfaction in managing home maintenance services.

1.5 System Features:

As a customer:

- 1- Create an account and log into the system.
- 2- Book maintenance services at suitable dates and times.
- 3- View and manage service bookings.
- 4- Track the status of service requests.
- 6- Rate technicians after service completion.

As a Technician:

- 1- Create and manage a professional account.
- 2- View and respond to customer service requests.
- 3- Manage appointments and work schedules.
- 4- Update the status of service requests.
- 5- View customer details for assigned jobs.
- 6- Manage personal profile and available services.

As an admin:

- 1- Manage customer and technician accounts.
- 2- Manage maintenance services and categories.
- 3- Monitor bookings and service requests.
- 4- Maintain the system and database.

1.6 Related Works (presenting previous works)

1- “ FIXORA” Website.

2- “ Urban Company ” Website.

1.7 System Requirements

The **Fixora** system can be accessed using a computer, laptop, tablet, or mobile device with an internet connection and a modern web browser. The minimum system requirements are:

- **Processor:** Dual-core processor or higher.
- **Memory:** Minimum 2 GB RAM.
- **Storage:** Sufficient storage for browser cache and temporary files.
- **Display:** Minimum screen resolution of **1024 × 768** pixels.
- **Internet Connection:** Broadband or Wi-Fi connection for accessing the web application and database.
- **Web Browser:** A modern browser such as Google Chrome, Microsoft Edge, Mozilla Firefox, or Safari.

These specifications ensure that the **Fixora** web application operates efficiently and provides a smooth user experience across different devices

1.8 System Users

1. **Customer:** Creates an account, browses maintenance services, books service requests, tracks bookings, and rates technicians.
2. **Technician:** Manages service requests, updates job status, manages appointments, and communicates with customers.
3. **Admin:** Manages customers, technicians, services, bookings, and the entire Fixora system.

1.9 Methodology

The **Fixora** system development process includes a set of activities, methods, tools, and best practices used to develop the web application's front-end efficiently.

1- Requirement Analysis:

The system requirements are collected and analyzed to understand the needs of customers, technicians, and administrators. These requirements include browsing services, booking appointments, managing user accounts, and viewing service requests.

2- System Design:

The design phase includes creating the website layout, designing the user interface, and organizing pages such as **Home, Services, About Us, Contact Us, Booking, Technician Profile, User Dashboard, and Login/Sign Up**.

3- Implementation:

The implementation phase focuses on developing the front-end of the application using **React.js, HTML, CSS, and JavaScript** to build a responsive and user-friendly interface.

3- Verification and Validation:

The application is tested to ensure that all pages, navigation, forms, and user interface components function correctly and provide a smooth user experience.

5- Maintenance and Improvement:

The system is continuously improved by fixing interface issues, enhancing performance, optimizing responsiveness, and adding new front-end features based on user feedback.

1.11 Structure of the report

Chapter One presents the project introduction, project idea and scope, problem definition, system objectives, system features, methodology, system requirements, and system users.

Chapter Two describes the system analysis, including functional and non-functional requirements, use case diagram, activity diagrams, sequence diagrams, and other analysis models.

Chapter Three presents the system design, including the class diagram, database design (ERD), and the user interface design of the **Fixora** web application.

Chapter Four describes the implementation process, including the technologies and tools used to develop the front-end of the system, as well as screenshots of the implemented user interfaces.

Chapter Five presents the testing process, including testing the user interface, navigation, responsiveness, and validation of the main system features.

Finally, **Chapter Six** provides the conclusion and future work by summarizing the project, discussing its achievements, identifying its limitations, and suggesting possible enhancements and additional features for future development.

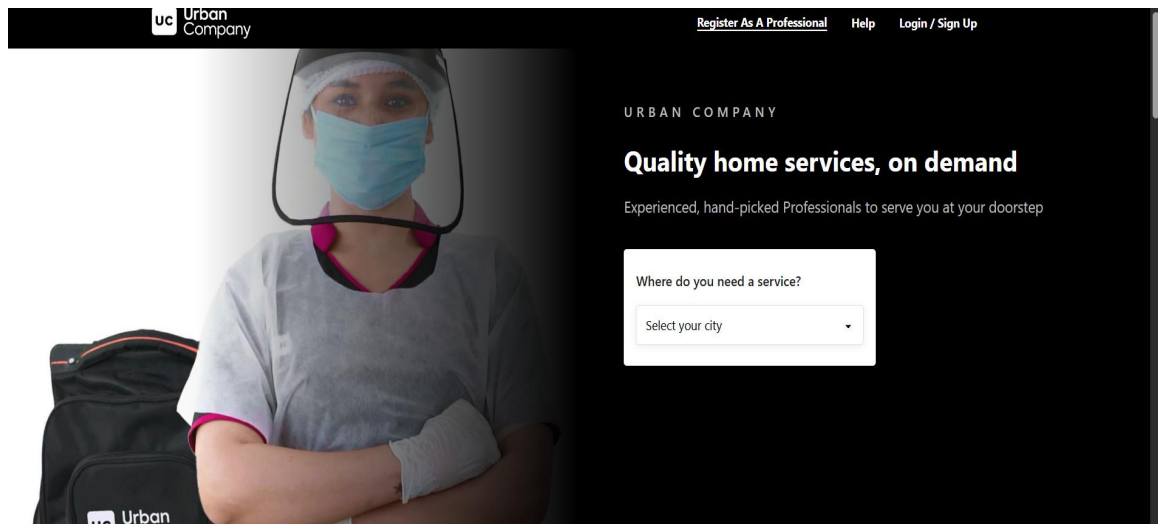
CHAPTER 2

Literature Review

Chapter 2: Literature Review

2-1 Presentation of existing solutions or similar projects:

“urban company ” Website



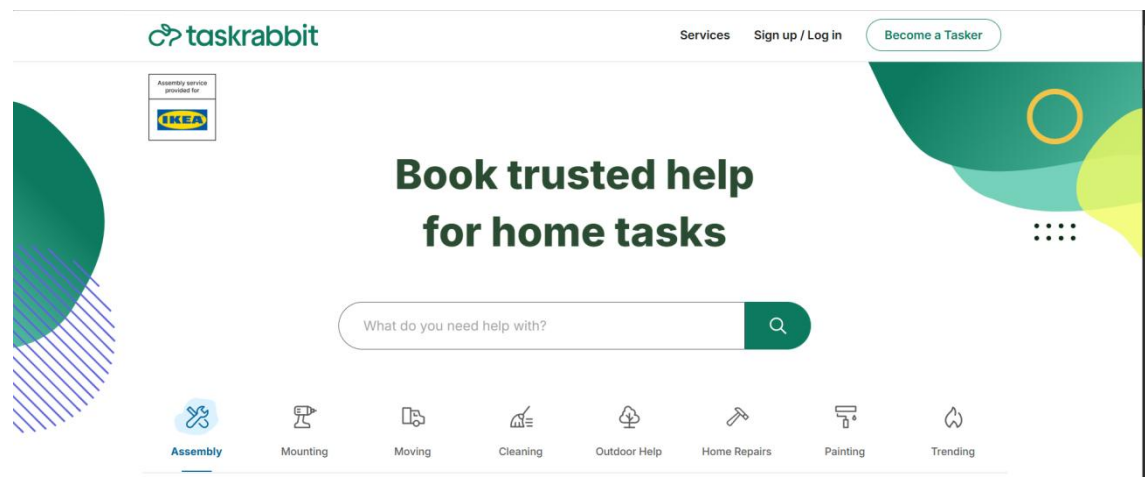
Advantages:

- 1- Provides a wide range of home maintenance services such as plumbing, electrical work, carpentry, painting, appliance repair, and cleaning.
- 2- Allows customers to book services online quickly and conveniently.
- 3- Offers verified and professional technicians with customer ratings and reviews.

Disadvantages:

- 1- Service availability may be limited in some locations.
- 2- Customers cannot always negotiate prices directly with technicians before booking.
- 3- Limited direct communication between customers and technicians before confirming the service request.

1- “TaskRabbit” Application



Advantages:

- 1- Users can register and log in to the application.
- 2- Provides a variety of home maintenance and repair services.
- 3- Allows customers to search for nearby skilled technicians.
- 4- Customers can book services online at convenient times.
- 5- Displays technician ratings and customer reviews.
- 6- Offers secure online payment options.
- 7- Allows users to track service requests.
- 8- Provides customer support through the platform.

Disadvantages:

- 1- Service availability is limited in some locations.
- 2- Prices may vary depending on the technician and demand.
- 3- Customers cannot always communicate directly with technicians before confirming a booking.
- 4- Some services include additional platform or service fees.

2-2 Compared with our project:

After checking multiple similar projects, our key advantages and what makes our project stand out are:

- 1- Easy and secure registration for **customers, technicians, and administrators**.
- 2- Customers can browse different maintenance services and choose the most suitable technician based on ratings and experience.
- 3- The platform provides a simple and transparent booking process with real-time service request tracking.
- 4- Technicians can manage appointments, update service status, and organize their work schedules efficiently.
- 5- Customers can rate and review technicians after service completion to improve service quality.
- 6- The system offers a simple, responsive, and user-friendly interface that can be accessed from different devices.
- 7- The admin can efficiently manage users, technicians, services, bookings, and monitor the overall operation of the platform from a centralized dashboard.

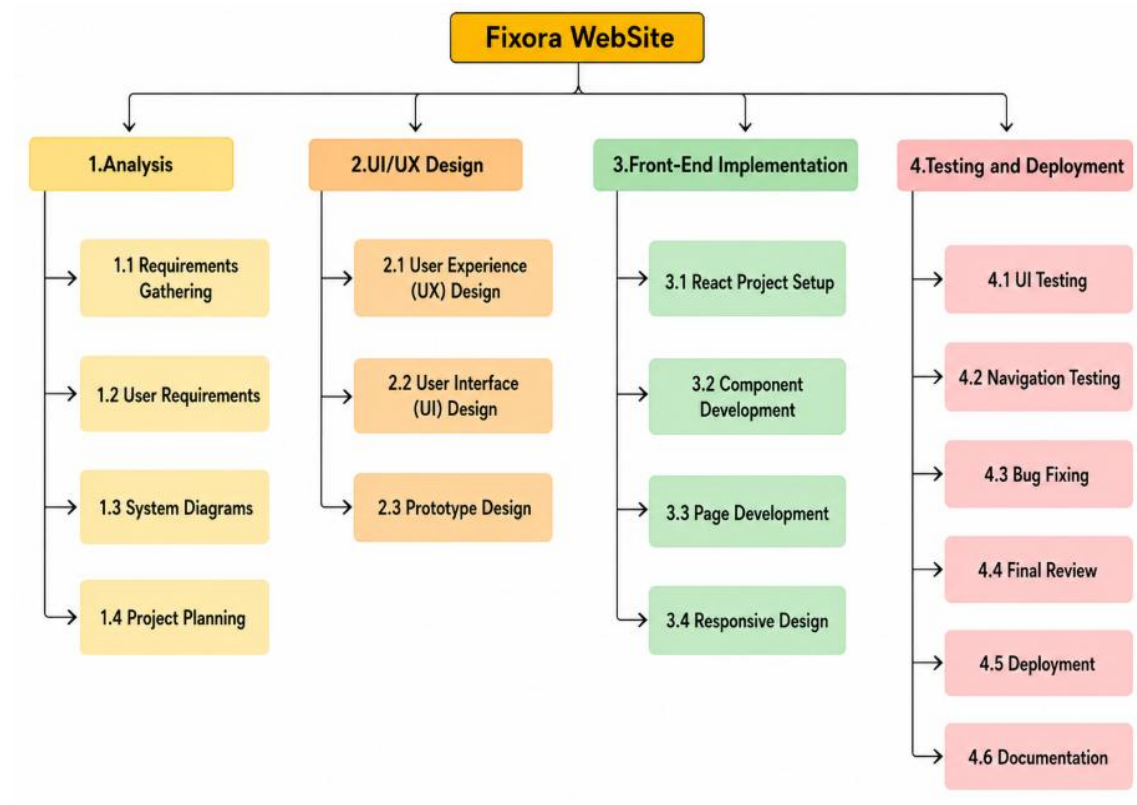
CHAPTER 3

System Analysis

Chapter 3: System Analysis:

3.1 Work Breakdown Structure (WBS)

The primary goal of using WBS was to break down the project into more digestible and manageable components. This method provided a clear framework for organizing the project, defining deliverables, and assigning responsibilities, ultimately enhancing project visibility and control. The WBS was implemented into our project as follows:



3.2 Requirements Analysis:

The functional and non-functional requirements of the application we have created will be detailed in the next section. These requirements have been combined to achieve the ultimate goal of the system to create an application where travelers and service owners who need or provide services will find it.

3.2.1 Functional requirements:

The following table will contain functional requirements, which can be defined as what the system should do and what the system should not do. They are the features of the final product and emphasize the requirements of the user.

Identifier	Requirement
Req-1	The system shall allow customers, technicians, and administrators to register a new account.
Req-2	The system shall allow users to log in using their email and password.
Req-3	The system shall allow users to reset their password if forgotten.
Req-4	The system shall allow users to view and update their personal profile.
Req-5	The system shall display all available maintenance service categories.
Req-6	The system shall allow customers to search for maintenance services or technicians.
Req-7	The system shall allow customers to book maintenance services.
Req-8	The system shall allow customers to view, update, or cancel service bookings.
Req-9	The system shall display technician profiles, including experience, skills, and ratings.

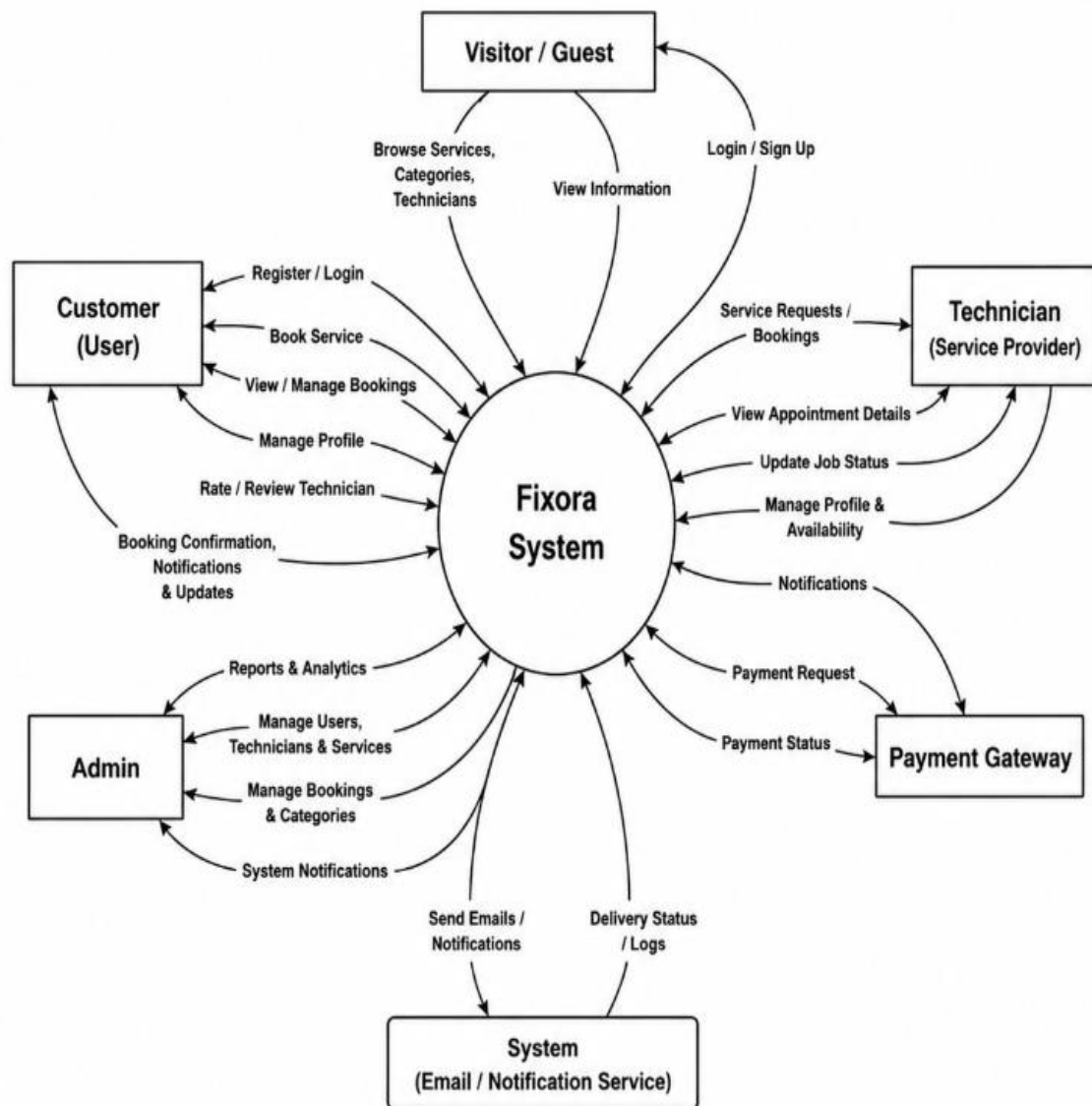
Req-10	The system shall allow technicians to accept or reject service requests.
Req-11	The system shall allow technicians to update the status of service requests (Pending, In Progress, Completed).
Req-12	The system shall allow technicians to manage their availability and appointments.
Req-13	The system shall allow customers to rate and review technicians after service completion.
Req-14	The system shall allow administrators to manage customers, technicians, and service categories.
Req-15	The system shall allow administrators to monitor all bookings and service requests.
Req-16	The system shall provide notifications for booking confirmations and service status updates.

3.2.2 Non-functional requirements

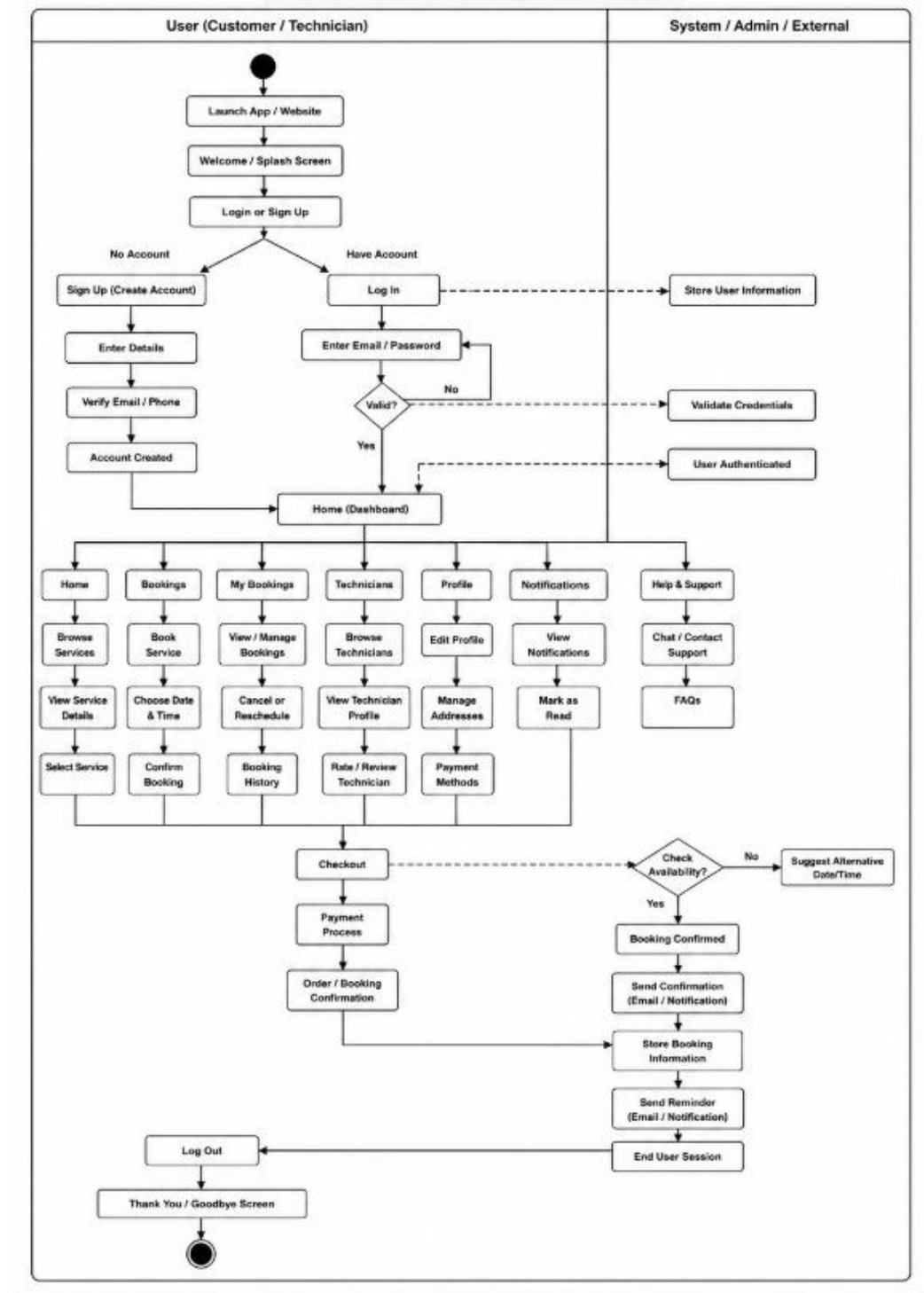
Identifier	Requirement
Req-1	The system shall load pages within 3 seconds under normal conditions.
Req-2	2. The system shall encrypt user passwords and sensitive data.
Req-3	The system shall be user-friendly and easy to navigate for all users.
Req-4	The system shall support increasing numbers of users without performance loss.
Req-5	The system shall be available 99% of the time except during maintenance.
Req-6	6. The system shall work on all modern browsers and mobile devices.
Req-7	The system shall be designed in a modular way to allow easy updates and fixes.

3.3 System Diagram:

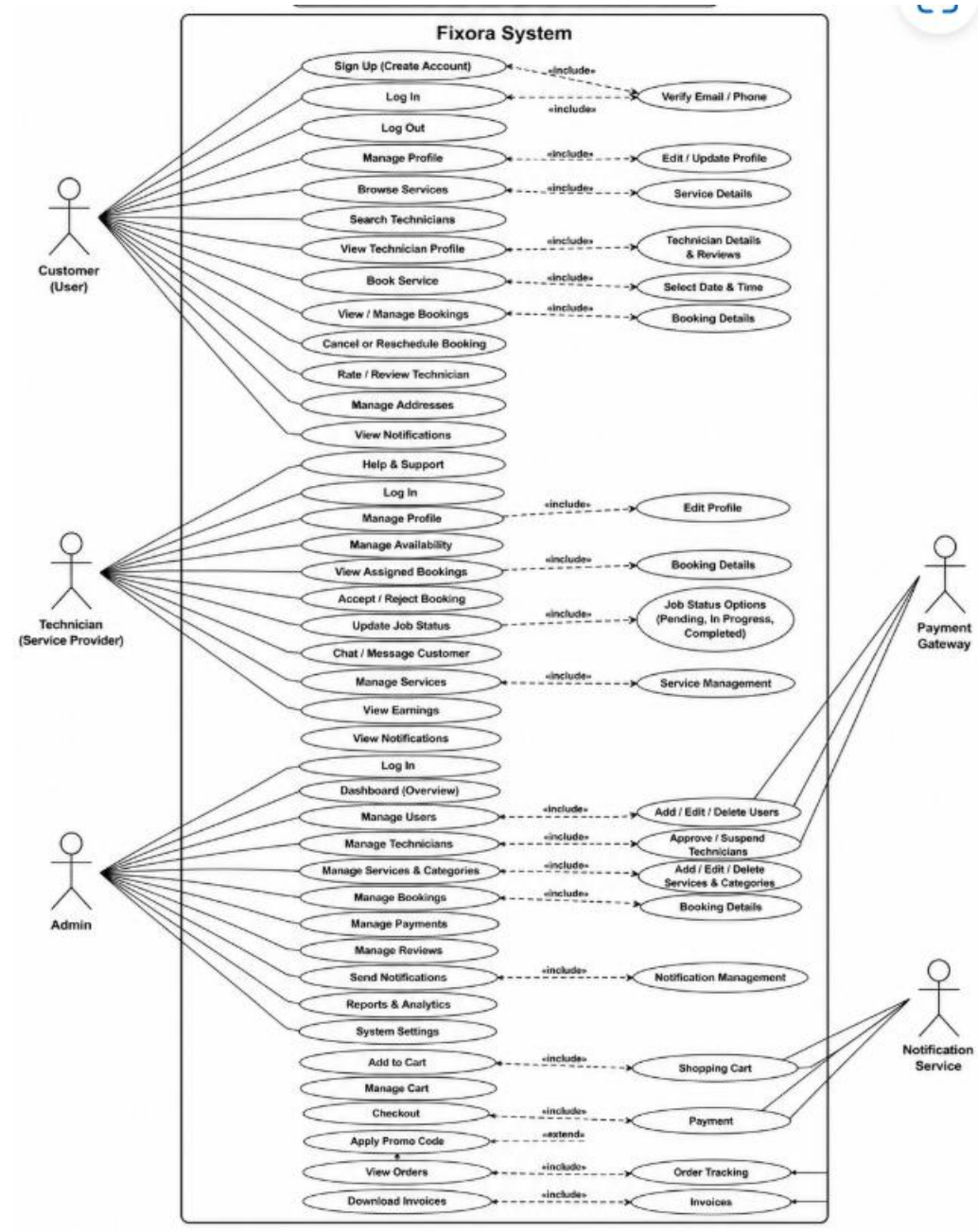
Context Diagram



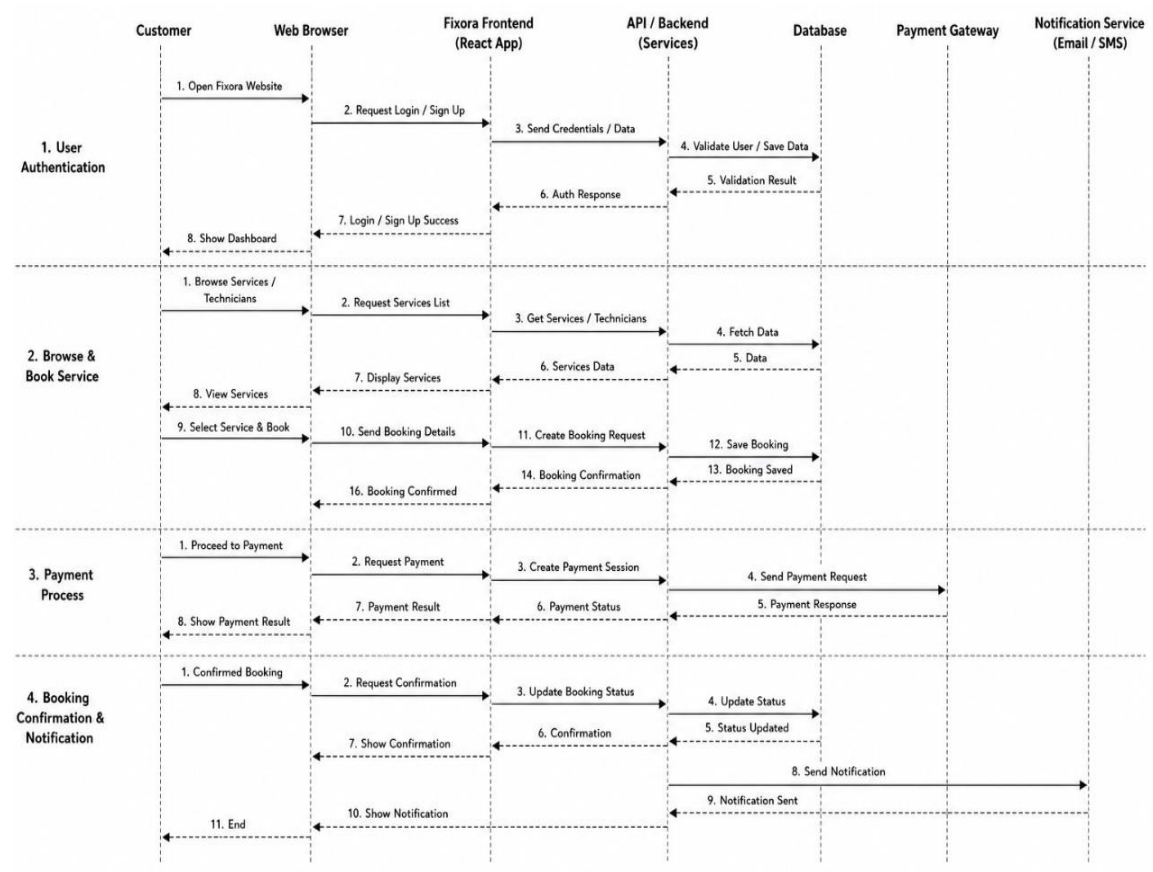
Activity Diagram



Use Case Diagram



Sequence Diagram

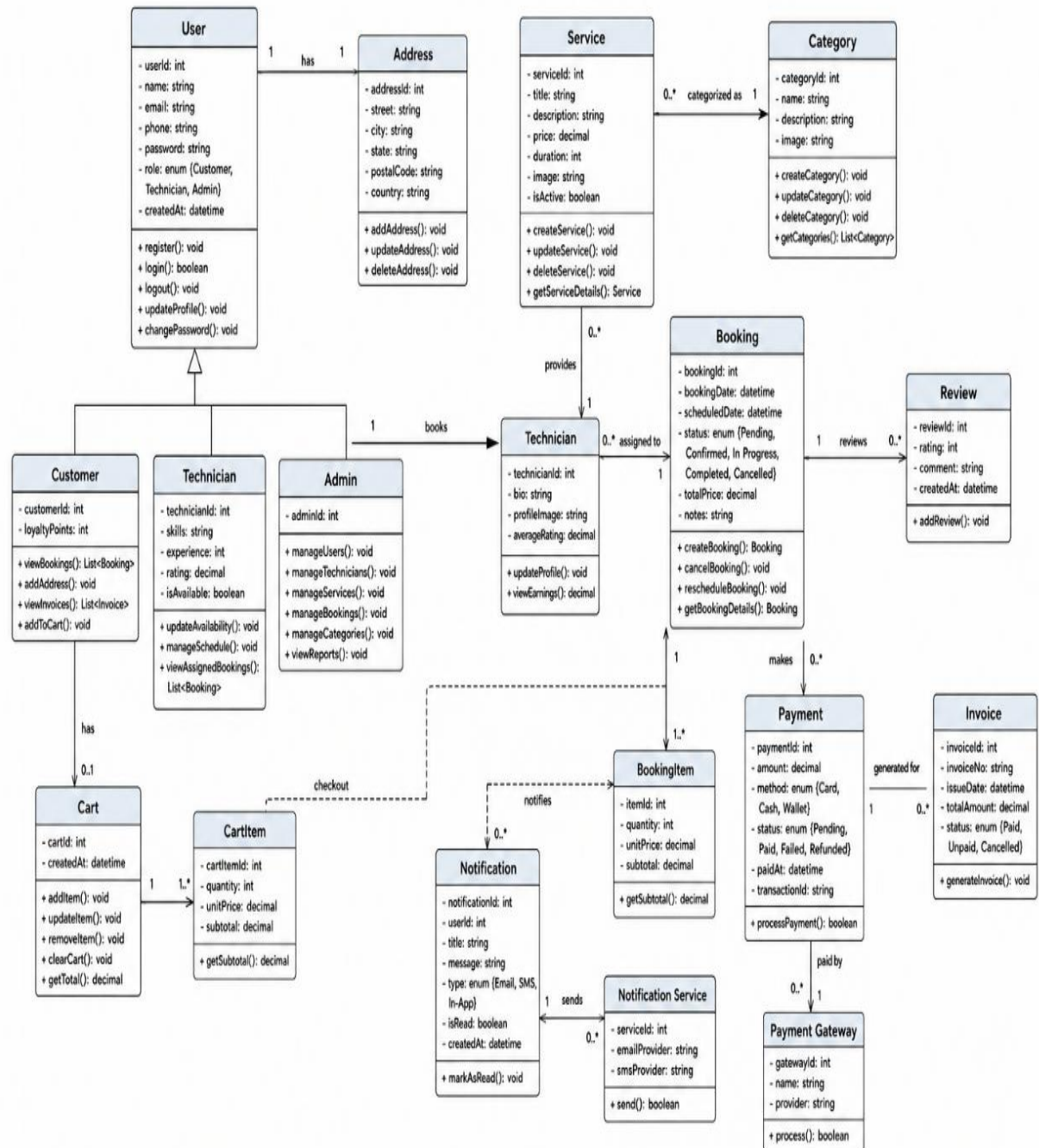


CHAPTER 4

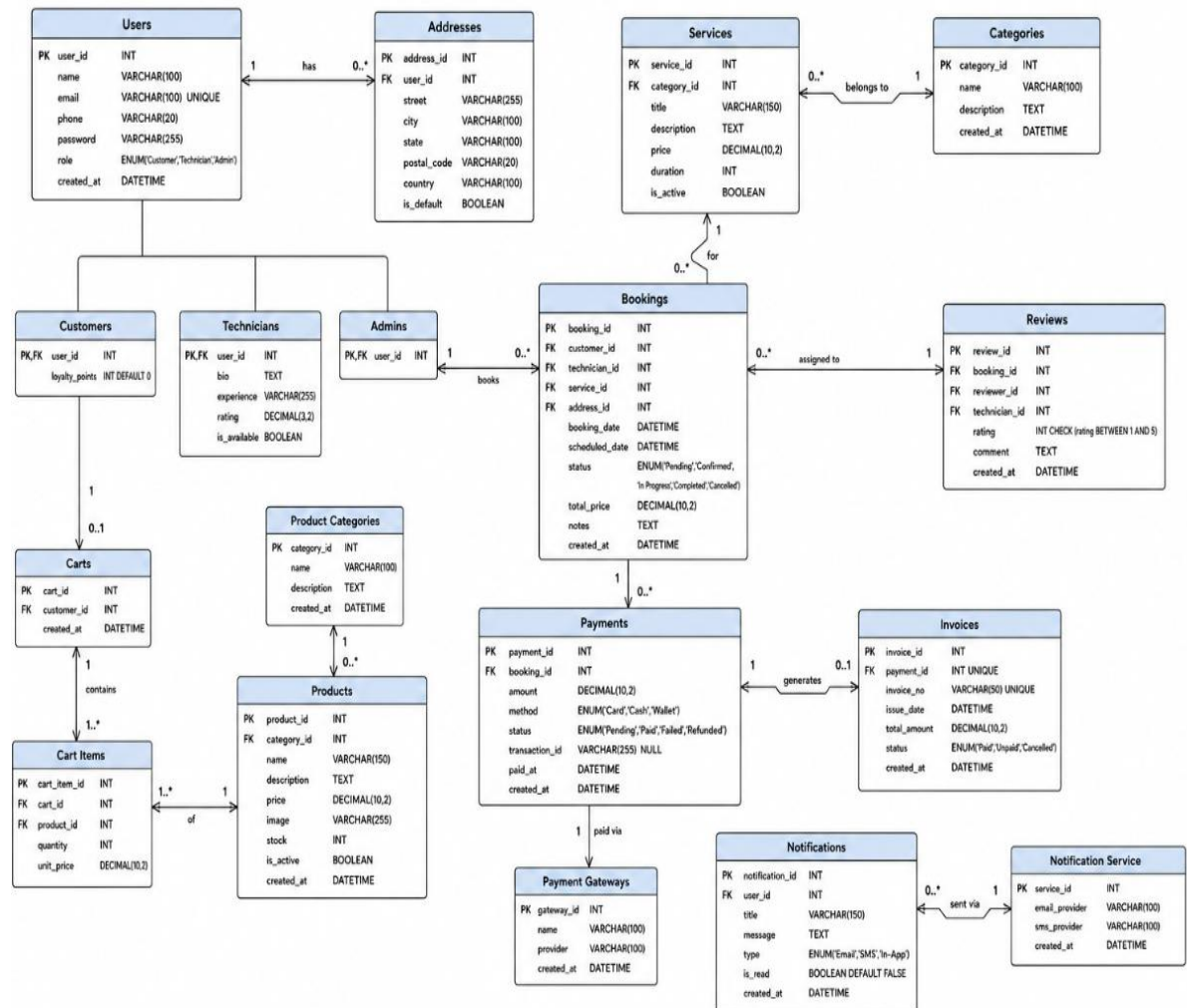
System Design

Chapter 4: System Design

4.1 Class Diagram



Data Base Structure ERD



CHAPTER 5

Implementation

Chapter 5:Implementation:

5.1 Tools used in the project:

React.js

React.js is a JavaScript library used to build the user interface of the **Fixora** website. It enables the development of reusable components and provides a fast, responsive, and interactive user experience.

HTML5

HTML5 was used to create the structure and content of the web pages, providing semantic elements that improve accessibility and maintainability.

CSS3

CSS3 was used to design and style the user interface, ensuring an attractive, responsive, and user-friendly layout across different screen sizes.

Java Script(ES6)

JavaScript was used to implement the interactive features of the website, such as form validation, navigation, and dynamic page behavior.

Git & GitHub

Git was used for version control, while GitHub was used to store the project repository, manage code versions, and facilitate collaboration during development.

FIGMA

Figma is used for UI/UX design, enabling collaborative design of the Animal Center's interface. It helps create prototypes and ensures that the final product is user-friendly and visually consistent.

Node . Js

Node.js is a JavaScript runtime environment used to manage the React development environment and execute project-related tools during development. It enables developers to install packages, run the development server, and build the application efficiently.

Tailwind

Tailwind CSS is a utility-first CSS framework used to design responsive and modern user interfaces. It provides pre-defined utility classes that simplify styling, improve development speed, and maintain a consistent design throughout the **Fixora** website.

npm

npm is the default package manager for Node.js. It is used to install and manage project dependencies, libraries, and development tools required for building and running the **Fixora** React application efficiently.

5.2 System Screenshots: SIGN in:



Welcome Back

Sign in to continue to Fixora

Email Address

 you@example.com

Password [Forgot password?](#)

☐ Remember me for 30 days

Login →

OR CONTINUE WITH

 Google

 Apple

SIGN up :



Join Fixora

Choose an account type to customize your experience



Homeowner

I'm looking for trusted professionals for home repairs and maintenance.

- ✓ Book in minutes
- ✓ Secure payments
- ✓ Review professionals



Professional

I offer professional services and want to reach new clients easily.

- ✓ Daily jobs
- ✓ Manage bookings
- ✓ Grow your business



Account



2
Details

Homeowner Account

Fill in your details to get started

Full Name

 John Doe

Email Address

 you@example.com

Phone Number

 +966 5X XXX XXXX

Identity Verification



ID Front Side



ID Back Side

Account

Details

Professional Account

Fill in your details and upload your ID to get started

Full Name

 John Doe

Email Address

 you@example.com

Phone Number

 +966

5X XXX XXXX

Your Skills

 Select your skills...

Identity Verification

HOME PAGE:



TRUSTED BY 50,000+ HOMEOWNERS

Welcome to Fixora

Find trusted professionals for every home service — quickly and confidently.

 Painting

Get Started →

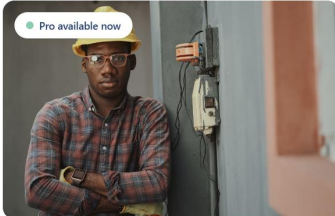
→ Login

Verified Professionals

Fast Booking

Secure Service

Pro available now



12K+ Professionals

98% Satisfaction

4.9 ★ Avg Rating

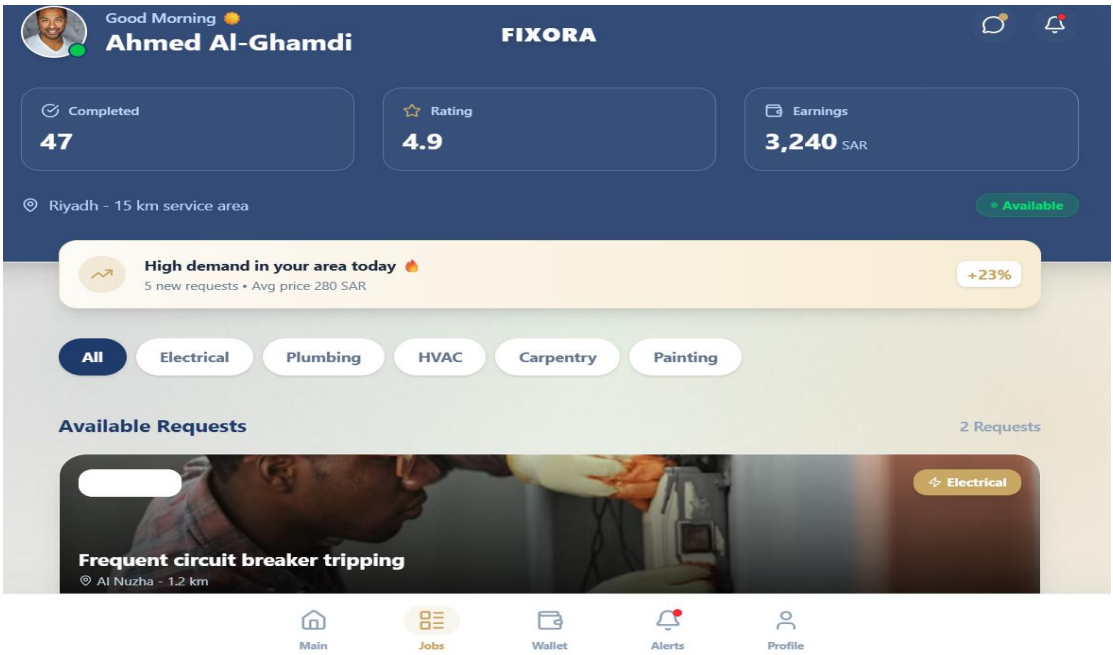
Electrical

Carpentry

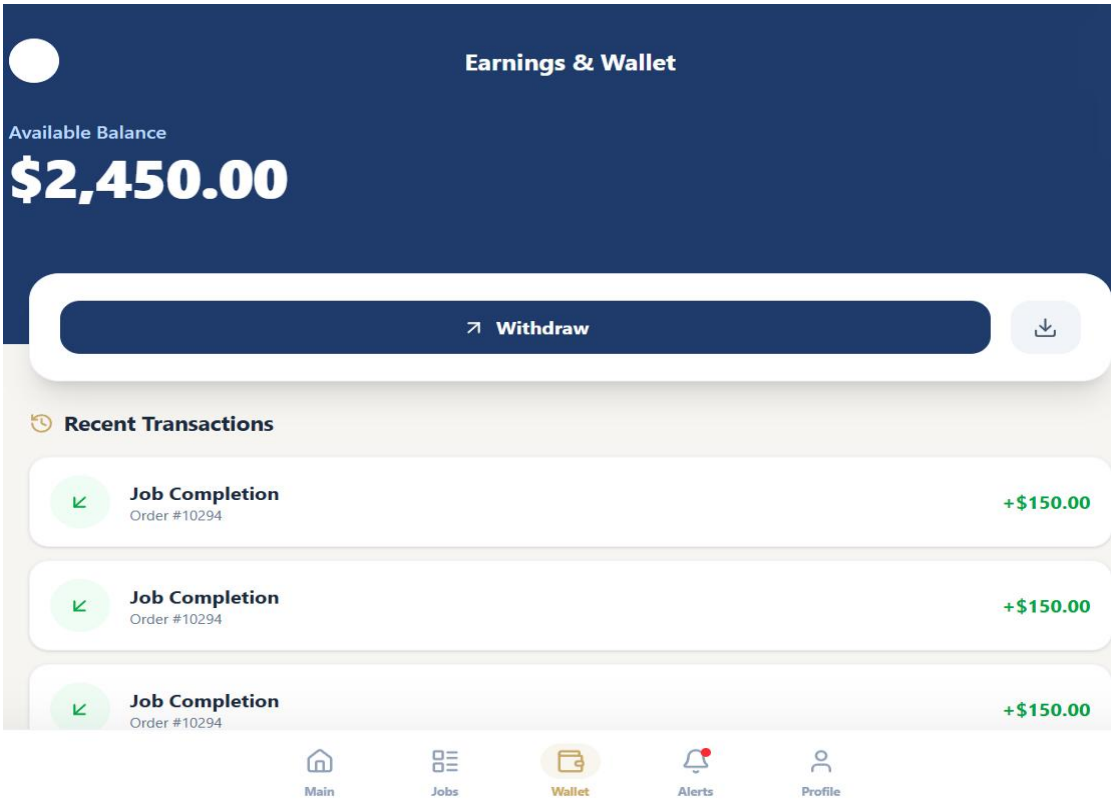
Repair

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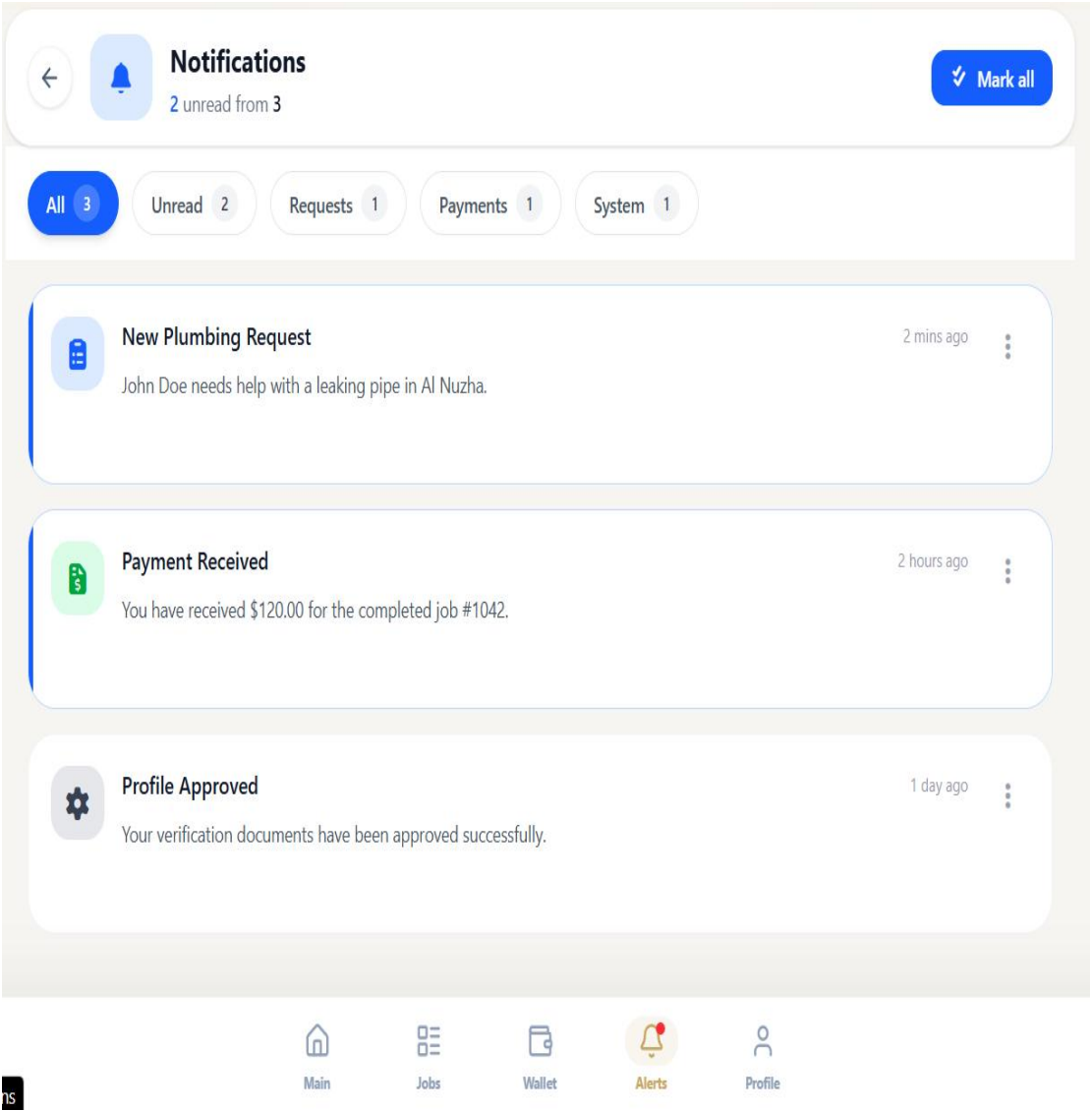
JOBS PAGE:



WALLET PAGE:



ALERTS PAGE:



CHAPTER 6

TESTING

Chapter 6 : TESTING :

Welcome Back

Sign in to continue to Fixora

Email Address

 erfcvn@gmail.com

Password

[Forgot password?](#)





☒ Remember me for 30 days

Login →

CHAPTER 7

Conclusion and Future Work

7.1 Conclusion:

We have developed **Fixora**, a web-based home maintenance service platform that connects customers with skilled technicians through a simple and user-friendly interface. The system allows users to register, create accounts, browse maintenance services, search for technicians, and book service appointments with ease.

The platform includes essential pages such as **Home, Services, About Us, Contact Us, Booking, Technician Profile, Login, and Sign Up**, providing a smooth and organized user experience. Customers can manage their bookings, while technicians can view and manage service requests. The administrator can monitor users, technicians, services, and bookings through the system.

Overall, **Fixora** provides an efficient, reliable, and centralized solution that simplifies the process of requesting and managing home maintenance services, improving convenience for both customers and technicians.

7.2 Future Work:

- 1- Develop **Android and iOS mobile applications** to make the platform more accessible and convenient for users.
- 2- Integrate **online payment gateways** to enable secure and seamless payments for maintenance services.
- 3- Add a **real-time chat and notification system** to improve communication between customers and technicians and provide instant service updates.
- 4- Integrate **AI-powered service recommendations** that suggest suitable maintenance services and technicians based on users' previous bookings and preferences.

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